

OJAS GRAMOPADHYE

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EDUCATION

New York University, New York Aug 2024 - May 2026
Master of Science, Computer Science *GPA : 3.93/4*
Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Algorithms, ML Systems & Operations

Indian Institute of Technology Bombay, India May 2019 - Jul 2023
Bachelor of Technology, Computer Science and Engineering *GPA : 7.74/10*
Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Data Structures, Software Systems

WORK EXPERIENCE

Verifiable Agent Infrastructure for Enterprise AI Governance Feb 2026 - Present
AI Engineering Intern **Arbis.AI**

- Building infrastructure for verifiable AI agents, focusing on auditability, governance, and secure decision workflows

Detection and Mitigation of Hallucination in Generative Models May 2023 - May 2024
Research Assistant **IBM Research | IIT Bombay**

- Improved diagnostic accuracy of LLMs to 90% by designing a Forward-Backward Chain-of-Thought evaluation pipeline and fine-tuning LLaMA-2 (7B, 70B) models for hallucination mitigation in medical QA
- Built a Subjective Medical QA benchmark and a reward-based verification pipeline that generates mixed-step reasoning signals and enables scalable, reproducible experimentation - EMNLP Findings 2024 (*arXiv: arxiv.org/abs/2403.04890*)

Japanese OCR Pipeline May 2022 - Jul 2022
Data Science Intern **Daikin**

- Built an OCR pipeline for 3,000+ Japanese documents, achieving levenshtein distance 0.91 using MobileNetV3 + DB
- Developed preprocessing and evaluation workflows for dataset preparation and performance measurement

Locust Occurrence Modeling and Prediction May 2021 - Jul 2021
Data Science Associate | R&D Team **Dtime.ai**

- Engineered spatiotemporal features from 40+ years of agroclimatic data and addressed class imbalance using SMOTE
- Improved locust attack precision by 10% via pixel-level classification using a ConvLSTM U-Net model

KEY PROJECTS

Flowsight : NYC Traffic Collision Analysis and Prediction Jan 2025 - May 2025
Course Project | Big Data **New York University**

- Built a Kafka-MongoDB-Spark pipeline over 5M+ NYC records, using speeds with crash context using haversine joins
- Trained a risk model where top 5% predicted grid cells captured 38% of collisions, surfaced in a React + D3 dashboard

Erica : AI Tutor agent with GraphRAG Sept 2025 - Dec 2025
Course Project | Artificial Intelligence **New York University**

- Built a Dockerized LLM tutor integrating GraphRAG with MongoDB-backed multi-format course content, constructing a typed knowledge graph for grounded, concept-aware retrieval
- Achieved MRR = 0.62 for retrieval while generating prerequisite-aware, citation-grounded explanations

Multi-subject alignment : Neural Speech Decoding Sept 2025 - Present
Graduate Research | NYU Video Lab **New York University**

- Improved cross-subject neural speech decoding by learning subject-invariant embeddings through contrastive alignment and adapter-based modeling across multiple ECoG participants
- Experimenting with representation learning by implementing a wav2vec2-based pretraining pipeline : masked encoder modeling, Gumbel quantization, negative sampling, and a Transformer context network for phoneme classification

TECHNICAL SKILLS

Machine Learning: LLMs, RAG systems, AI agents, NLP, deep learning, contrastive learning, representation learning
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, LangGraph, LangChain, Accelerate, DeepSpeed
Data & Systems: Apache Kafka, Apache Spark, Dask, MongoDB, PostgreSQL, ChromaDB, MLflow, Docker, Linux, Bash
Software Engineering: Python, C++, JavaScript, TypeScript, FastAPI, React, Streamlit, d3.js, Git, REST APIs